

Exercises for Hands-on-session in DFT-FE Workshop

(1) O₂ molecule ground-state calculations (non-periodic system)

Detailed Problem statement:

In this exercise, you will compute the bond length of an O₂ molecule in its triplet spin configuration using LDA, GGA, and MGGA exchange–correlation (XC) functionals. First, perform a convergence study to determine the optimal mesh parameters for your calculations. Once the mesh settings are established, carry out ion-relaxation (geometry optimization) calculations for the O₂ molecule using each of the three XC functionals. The results should report the equilibrium bond length obtained with LDA, GGA, and MGGA XC functionals.

Step-1: Mesh convergence study

- Go to the folder **“Example1/Converge”**
- In each XC-functional subfolder (**“LDAPW,” “PBE,” and “R2SCAN”**), locate the parameter files for every mesh-size and polynomial-order combination.
- Use the provided **“referenceMesh.prm”** and **“referenceOutput”** files as references for energies and forces.
- Run the jobs for each functional and fill in the table below with your results.
- For reference, example output files are available in **“Example1/Converge/<XC_functional>/out.”**

Mesh	DoFs	Free Energy in Ha	Atom 0 Force in Ha/bohr	Diff in energy/atom wrt reference	Diff in Atom 0 force wrt. reference
P = 6, h = 1.4					
P = 6, h = 1.2					
P = 6, h = 1.0					
p = 7, h = 1.4					
P = 7, h = 1.2					

Step-2: Density Visualisation

- Go to the **“Example1/Visualize/”** folder.
- Run the job using the parameter file provided in the folder; upon successful completion, a **“densityOutputFolder”** will be created.
- To visualize the results in ParaView, open the master **“.pvtu”** file.
- For reference, example output files are available in **“Example1/Visualize/out.”**

Step-3 Ion Relaxation:

- Go to the **“Example1/Relax”** folder.
- Navigate to the subfolder for each XC functional (**“LDAPW,” “PBE,” or “R2SCAN”**).
- In the corresponding parameter file in each subfolder, note the following settings.
“SOLVER MODE = GEOOPT”
“SPIN POLARIZATION = 1”
- The 6th column in **“coordinates.inp”** specifies the initial guess of magnetization of each atom.
- Run the jobs for each functional and complete the table provided below.

XC Functional	Bond Length (Bohr)	Free energy (Ha)	Net Magnetization (μ_B)
LDA-PW			
GGA-PBE			
MGGA-R2SCAN			

(2) Phosphorene Monolayer with a vacancy (semi-periodic system)

In this exercise, you will calculate the vacancy formation energy of monolayer phosphorene containing a single vacancy using the DFT-FE code. You will begin by computing the ground-state energy of fully relaxed pristine phosphorene, followed by determining the ground-state energy of phosphorene with a single atomic vacancy after relaxing the ionic positions. These two energies will then be used to evaluate the vacancy formation energy. The simulation domain is semi-periodic-periodic in the in-plane directions and non-periodic in the out-of-plane direction-and all calculations employ the PBE exchange–correlation functional. The detailed procedure for this calculation is provided below.

step-1: Evaluation of ground state energy of the relaxed mono-layer phosphorene

- Navigate to the “**Example2/pristine/**” directory.
- The relaxed pristine structure coordinates are provided in “**coordinates.inp**”, and the domain vectors are in “**domainVectors.inp.**”
- In “**p_scf.prm**”, the boundary conditions are specified with “**PERIODIC1 = true,**” “**PERIODIC2 = true,**” and “**PERIODIC3 = false.**”
- The calculation uses a converged finite-element mesh with:
polynomial order = 5
mesh size ≈ 1.4
atom ball radius = 4 Bohr

step-2: Ionic relaxation of mono-layer phosphorene with single atom vacancy

- Navigate to the “**Example2/withVac/**” directory.
- A single phosphorus atom has been removed from the center of the phosphorene sheet to create a vacancy; the supercell is chosen large enough to avoid interactions between periodic images of the vacancy.
- In the “**p_geoOpt.prm**” file, note the following settings:
“**SOLVER MODE = GEOOPT**”
“**OPTIMIZATION MODE = ION**”
“**FORCE TOL = 5e-4**”
Boundary conditions: “**PERIODIC1 = true**” “**PERIODIC2 = true**” “**PERIODIC3 = false**”
- The converged finite-element mesh uses:
polynomial order = 5
mesh size ≈ 1.4
atom ball radius = 4 Bohr

Vacancy formation energy = (energy of the system with vacancy) + $((1/N) - 1)$ * (energy of the pristine phosphorene) (N: Number of phosphorous atoms in the pristine phosphorene.)

Run the jobs and complete the below table.

System	Free energy (Ha)
Pristine phosphorene	
Phosphorene with vacancy	

(3) Silicon supercell with a vacancy (fully periodic system)

In this exercise, you will calculate the vacancy formation energy of Si vacancy using 5x5x5 supercell (1000 atoms) using the DFT-FE code with PAW formalism. First, you will determine the ground-state energy of pristine Silicon supercell. Next, you will compute the ground-state energy of Silicon supercell containing a single atomic vacancy after relaxing the ionic positions. Using these two energies, you will then evaluate the vacancy formation energy.

In this example, the simulation domain is **fully periodic**: periodic in all directions. The PBE exchange–correlation functional is employed throughout, and you will employ PAW formalism to do this calculation. The detailed procedure for this calculation is described below.

step-1: Evaluation of ground state energy of the silicon supercell (no defect)

Navigate to the “**Example3/GS**” directory. The coordinates of the relaxed pristine structure and the corresponding domain vectors for the pristine Silicon are given in the files “coordinates.inp” and “domainVectors.inp”, respectively.

- In the file “parameterFile-GS.prm”, notice the following settings
 - “**CELL STRESS = true**” (for evaluation of cell stress)
 - Boundary conditions: “**PERIODIC1 = true**” “**PERIODIC2 = true**” “**PERIODIC3 = true**”
 - The converged finite-element mesh: “**POLYNOMIAL ORDER = 4, MESH SIZE AROUND ATOM = 1.8**”

step-2: Ionic relaxation of Silicon supercell with single atom vacancy

Navigate to the “**Example3/RLX**” directory. A single silicon atom is removed from the center of the Silicon supercell to create a vacancy. The supercell is chosen to be sufficiently large to eliminate interactions between periodic images of the vacancy.

- In the file “parameterFile-GS.prm”, notice the following settings,
 - “**SOLVER MODE = GEOOPT**”
 - “**OPTIMIZATION MODE = ION**”
 - “**CELL STRESS = true**”
 - Boundary conditions: “**PERIODIC1 = true**” “**PERIODIC2 = true**” “**PERIODIC3 = true**”
 - The converged finite-element mesh: “**POLYNOMIAL ORDER = 4, MESH SIZE AROUND ATOM = 1.8**”

Run the jobs and complete the below table.

Vacancy formation energy = (energy of the system with vacancy) + $((1/N) - 1)$ * (energy of the pristine system)

(N: Number of Silicon atoms in the pristine silicon supercells.)

System	Free energy (Ha)
Pristine Silicon supercell	
Silicon with vacancy	

(4) Use cases: Non-collinear + SOC

In this exercise, you will calculate the magnetic anisotropy energy of FePt nanoparticle (30 atoms) using the DFT-FE code. First, you will determine the ground-state energy of relaxed nanoparticle using a spin-polarized ground state calculation. Next, you will use the computed ground state charge and magnetization densities and perform NSCF calculations accounting for non-collinear magnetization and Spin-Orbit-Coupling (SOC).

In this example, the simulation domain is **fully non-periodic**. The PBE XC functional is employed throughout, and you will employ scalar-relativistic and fully relativistic ONCV pseudopotentials as appropriate. The detailed procedure for this calculation is described below.

step-1: Evaluation of ground state densities of the relaxed FePt nanoparticle

Navigate to the “**Example4/**” directory. The coordinates of the relaxed structure and the corresponding domain vectors for the nanoparticle are given in the files “coordinates.inp” and “domainVectors.inp”, respectively.

- In the file “parameterFile.prm”, notice the following settings
 - “**SOLVER MODE = GS**”.
 - Boundary conditions: “**PERIODIC1 = false**” “**PERIODIC2 = false**” “**PERIODIC3 = false**”
 - “**SPIN POLARIZATION = 1**”, “**NONCOLLINEAR SPIN = false**”, “**SPIN-ORBIT COUPLING = false**”
 - The converged finite-element mesh: “**POLYNOMIAL ORDER = 6**, **MESH SIZE AROUND ATOM = 0.8**” and “**ATOM BALL RADIUS = 4**”
 - “**SAVE QUAD DATA = true**” (to save the ground state densities)

step-2: Evaluation of band energies for various spin orientations

Navigate to the “**Example4/**” directory. Edit the file “parameterFileNSCF.prm” as follows.

- In the file “parameterFile.prm”, notice the following settings
 - “**SOLVER MODE = NSCF**”.
 - Boundary conditions: “**PERIODIC1 = false**” “**PERIODIC2 = false**” “**PERIODIC3 = false**”
 - “**SPIN POLARIZATION = 1**”, “**NONCOLLINEAR SPIN = false**”, “**SPIN-ORBIT COUPLING = false**”
 - The converged finite-element mesh: “**POLYNOMIAL ORDER = 6**, **MESH SIZE AROUND ATOM = 0.8**” and “**ATOM BALL RADIUS = 4**”
 - “**LOAD QUAD DATA = true**” (To load the previously saved density data)
 - “**MAG PHI = 0**”, “**MAG THETA = 0**”
 - Repeat for various spin orientations.

SOC	E_x	E_z	MAE (E_x - E_z)
OFF			
ON			

Example 5: Strong Scaling of the DFT-FE Code Using a 5×5×5 Silicon Supercell

In this example, you will evaluate the **strong scaling performance of the DFT-FE code** using a fixed 5×5×5 silicon supercell (1000 atoms). The problem size remains constant, and you will vary the number of GPUs to measure how the wall time changes as more computational resources are added.

Navigate to “**Example5/**” directory. You will run the same input using **4, 8, 16, 24, 32, 40, and 48 GPUs** on Siddhi AI (each node contains 8 NVIDIA A100 GPUs). For each run, the maximum SCF iterations have been set to **5**.

For your reference find job scripts in “Example5/jobs” folder. Notice that for we have one task per GPU, i.e. ntasks-per-node is same as number of GPUs requested.

Your task is to record:

- **DoFs/GPU** (total DoFs divided by number of GPUs)
- **Wall Time** for the 5 SCF iterations
- **Scaling Efficiency**, computed as:

$$\text{Scaling Efficiency} = T_4/T_N \times (4/N)$$

# GPUs	DoFs/GPU	Wall Time	Scaling Efficiency
4			
8			
16			
24			
32			
40			
48			